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CLINICAL GUIDELINE

Stroke Rehabilitation: Synopsis of the 2024 U.S. Department of Veterans Affairs and U.S. Department of Defense Clinical Practice Guidelines

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Description: In July 2024, the U.S. Department of Veterans Affairs (VA) and U.S. Department of Defense (DOD) released a joint update of their 2019 clinical practice guideline (CPG) for the management of stroke rehabilitation. This synopsis is a condensed version of the 2024 CPG, highlighting the key aspects of the guideline development process and describing the major recommendations.

Methods: The VA/DOD Evidence-Based Practice Work Group convened a joint VA/DOD guideline development work group (WG) that included clinical stakeholders and conformed to the Institute of Medicine's tenets for trustworthy CPGs. The guideline WG conducted a patient focus group, developed key questions, and systematically searched and evaluated the literature (English-language publications from 1 July 2018 to 2 May 2023). The GRADE (Grading of Recommendations Assessment, Development and Evaluation) system was used to evaluate the evidence. The WG developed 47 recommendations along with algorithms for stroke rehabilitation in the inpatient and outpatient settings. Stakeholders outside the WG reviewed the CPG before approval by the VA/DOD Evidence-Based Practice Work Group.

Recommendations: This synopsis summarizes where evidence is strongest to support guidelines in crucial areas relevant to primary care physicians: transition to community (case management, psychosocial or behavioral interventions); motor therapy (task-specific practice, mirror therapy, rhythmic auditory stimulation, electrical stimulation, botulinum toxin for spasticity); dysphagia, aphasia, and cognition (chin tuck against resistance, respiratory muscle strength training); and mental health (selective serotonin reuptake inhibitor use, psychotherapy, mindfulness-based therapies for treatment but not prevention of depression).

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Stroke is a major cause of morbidity, mortality, and disability worldwide. It is a pervasive medical condition affecting nearly 800 000 persons annually in the United States; approximately 75% of cases are first-time occurrences and the remaining 25% are recurrent strokes (1). Roughly 3% of the U.S. population has experienced a stroke, with a projected increase to 4% by 2030 (1). Stroke is the fifth most prevalent cause of death in the United States, accounting for 1 out of every 21 deaths in the nation; alarmingly, a stroke-related death occurs approximately every 3 minutes 17 seconds (1). Stroke is a leading contributor to long-term disability, with approximately 45% of persons aged 15 to 50 years having at least moderate disability after a stroke (2).

The spectrum of disability resulting from stroke manifests diversely. Typical presentations may include motor weakness and sensory disturbances, impairments in speech and swallowing, vision loss or neglect, cognitive challenges involving inattention or memory loss, and emotional difficulties, such as mood disorders or anxiety. Stroke survivors consequently require tailored and timely rehabilitative interventions aligning with their individualized needs (3, 4). Tailored rehabilitative efforts should start as soon as clinically feasible to maximize functional outcomes. These guidelines are relevant for health care professionals across a stroke patient's continuum of care, including primary care providers, specialists (physiatry, neurology, and cardiology), nurses, and allied health professionals.

The patient population of interest for the 2024 VA/DOD Clinical Practice Guideline for Management of Stroke Rehabilitation (5) is adult patients with post-stroke deficits (motor, cognitive, speech, or sensory) who are candidates for rehabilitation. These guidelines can be found at www.healthquality.va.gov/guidelines/rehab/stroke/index.asp.

The focus and scope of the 2024 clinical practice guideline (CPG) is to provide primary care providers in the U.S. Department of Veterans Affairs (VA) and U.S. Department of Defense (DOD) with recommendations and tools for the rehabilitation management of adult patients (aged ≥18 years) who have had a stroke, with

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an emphasis on an interdisciplinary care approach. The guideline also provides stroke rehabilitation specialist providers with guidelines for evidence-based practice. It is intended to improve quality of care and clinical outcomes; however, it is not intended to define a standard of care. The 2024 guideline is an update to the 2019 VA/DOD Clinical Practice Guideline for Management of Stroke Rehabilitation (6).

Guideline Development Process

The VA/DOD developed the CPG through a process outlined by the VA/DOD Evidence-Based Practice Work Group (7, 8). Four guideline panel cochairs, 2 from the VA and 2 from the DOD, were chosen. Leadership from the VA and DOD then selected a multidisciplinary panel of practicing clinician stakeholders with specialists from various disciplines (**Supplement Table 1**, available at Annals.org). All members of the guideline work group (WG) completed conflict-of-interest disclosure forms for relationships in the previous 2 years and affirmed the disclosures verbally throughout the duration of the project. No conflicts of interest were identified. Leadership from the VA and DOD conducted a virtual patient focus group aimed at gaining insights into patients who are currently or were receiving stroke rehabilitation. Patient focus group participants were given the opportunity to review the final draft of this CPG and provide additional feedback.

The WG developed 12 key questions using the PICOTS (population, intervention, comparator, outcomes, timing of outcomes measurement, and setting) framework (**Supplement Table 2**, available at Annals.org). Key questions focused on rehabilitation interventions in the following areas: motor; cognitive, speech, or language; dysphagia; spasticity; mental or behavioral health; sensory rehabilitation; neglect; technology-assisted tools; settings and models of care; and caregiver support. A systematic search of peer-reviewed, English-language literature published from 1 July 2018 (the end date for the 2019 CPG evidence review) to 2 May 2023 was undertaken to address the key questions. Systematic reviews (SRs) and meta-analyses (MAs) were prioritized and supplemented with individual randomized controlled trials (RCTs) when appropriate. In general, inclusion criteria required that at least 80% of patients in an included study were adults (aged ≥18 years) who had had a stroke. Search methods and results can be found in the full CPG (5) and the **Supplement Figure** (available at Annals.org).

The GRADE (Grading of Recommendations Assessment, Development and Evaluation) method was used to evaluate the quality of the evidence and the strength of the recommendations (9–11). In the GRADE system, the ultimate strength of a recommendation is based on the lowest quality-of-evidence rating for any critical outcome in the body of literature being reviewed for a particular recommendation (Developing Evidence-Based Recommendations in the **Supplement**, available at Annals.org). Using the GRADE method, we assigned each recommendation a strength of "strong for," "weak for," "neither for nor against," "weak against," or "strong against."

Highlighted 2024 Guideline Updates

Six key areas explored in the 2024 CPG full guideline are 1) transitions to community; 2) motor therapy; 3) dysphagia, aphasia, and cognition; 4) mental health; 5) telehealth; and 6) noninvasive brain stimulation (**Supplement Table 2**). In this synopsis of the full guideline, recommendations with stronger evidence are highlighted. The following text includes comparisons to the 2019 CPG recommendations.

Key Area 1: Transition to Community

Patients with stroke and their caregivers may have significant challenges in negotiating the health care system to obtain the care they need. Evidence supports that case management services at the time of discharge from the acute care hospital or postacute care facility improve activities of daily living (ADLs) and functional independence of patients with stroke (new recommendation; "weak for"; recommendation 1) (**Table**) (12–225). Case management, compared with usual care, had a modest positive effect on ADLs and a small positive effect on mental health (12). Patients, caregivers, and providers overwhelmingly value case management.

Evidence also supports that behavioral health or psychosocial interventions improve patient and caregiver depression and that psychoeducation improves family function, patient functional independence, and quality of life (new recommendation; "weak for"; recommendation 2) (**Table**). Behavioral health or psychosocial interventions include supportive therapies, such as stress or coping management, motivational interviewing, and other similar supportive therapies (14–17). Psychoeducation might include information on stroke care or illness self-management, educational support and counseling, skills training, and use of social resources. Four SR and MAs including a total of more than 10 000 patients provided evidence that these types of interventions are beneficial (14–17).

Key Area 2: Motor Therapy

The primary care provider can recognize poststroke motor deficits and refer the patient for appropriate motor therapies, often delivered by physical or occupational therapy. The recommendations and tools for this key area are highlighted here.

General Motor Therapy

Evidence supports the use of task-specific practice to improve motor function, gait, posture, and ADLs (no change in recommendation; "strong for"; recommendation 5) (**Table**). Task-specific practice (also known as task-oriented practice or repetitive task practice) is used for improving motor function, gait, posture, and

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Table. VA/DOD Clinical Practice Guidelines on Stroke Rehabilitation and Supporting Evidence

Guideline Number	2019 VA/DOD Guideline Recommendation (Strength)	2024 VA/DOD Guideline Recommendation	2024 Strength of Recommendation	2024 Category	2024 References
Transitions to community					
1	We recommend a team-based approach in an organized inpatient unit that encompasses comprehensive rehabilitation in order to improve likelihood of discharge to home after acute stroke. (Strong for)	We suggest using case management services at time of discharge from the acute care hospital or post-acute care facility to improve activities of daily living and functional independence.	Weak for	Reviewed, new-added	Supporting (12) Additional (13)
2	–	We suggest the following interventions for patients and their caregivers Behavioral health/psychosocial interventions to improve patient and caregiver depression Psychoeducation to improve family function, patient functional independence, and quality of life	Weak for	Reviewed, new-added	Supporting (14–17)
3	–	There is insufficient evidence to recommend for or against implementing transitional care rehabilitation interventions (e.g., home-based services after hospital discharge) or early supported discharge to improve activities of daily living or functional disability following stroke.	Neither for nor against	Reviewed, new-replaced	Supporting (18–22)
4	–	There is insufficient evidence to recommend for or against community participation interventions to improve community engagement for survivors of stroke.	Neither for nor against	Reviewed, new-added	Supporting (23)
–	We recommend that rehabilitation therapy should start as soon as medical stability is reached. (Strong for)	–	–	–	–
–	There is insufficient evidence to recommend for or against implementing very early mobilization (within 24–48 h) to improve functional outcomes. (Neither for nor against)	–	–	–	–
–	There is insufficient evidence to recommend for or against early supported discharge. (Neither for nor against)	–	–	–	–
Motor therapy					
5	We recommend task-specific practice (also known as task oriented practice or repetitive task practice) for improving upper and lower extremity motor function, gait, posture, and activities of daily living. (Strong for)	We recommend task-specific practice (also known as task-oriented practice or repetitive task practice) to improve motor function, gait, posture, and activities of daily living.	Strong for	Reviewed, not changed	Supporting (24–34)
6	–	We suggest mirror therapy to improve motor outcomes and activities of daily living.	Weak for	Reviewed, new-replaced	Supporting (35–38) Additional (39, 40)
7	There is insufficient evidence to recommend for or against mirror therapy for improvements in limb function. (Neither for nor against)	We suggest mirror therapy to improve unilateral spatial neglect.	Weak for	Reviewed, new-added	Supporting (41) Additional (39, 40, 42–45)

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Guideline Number	2019 VA/DOD Guideline Recommendation (Strength)	2024 VA/DOD Guideline Recommendation	2024 Strength of Recommendation	2024 Category	2024 References
8	We suggest offering body-weight support treadmill training as an adjunct to gait training in the non-ambulatory patient. (Weak for)	There is insufficient evidence to recommend for or against body-weight support treadmill training to improve motor outcomes.	Neither for nor against	Reviewed, new-replaced	Supporting (46–49)
9	We suggest offering rhythmic auditory cueing as a modality to include in multimodal interventions to improve walking speed. (Weak for)	We suggest rhythmic auditory stimulation as an adjunct intervention to improve motor outcomes.	Weak for	Reviewed, new-replaced	Supporting (50–54)
10	–	There is insufficient evidence to recommend for or against the use of high intensity interval training over moderate intensity continuous training to enhance gait recovery.	Neither for nor against	Reviewed, new-replaced	Supporting (55–57) Additional (58, 59)
11	We suggest offering Constraint-Induced Movement Therapy or modified Constraint-Induced Movement Therapy for individuals with at least 10 degrees of active extension in two fingers, the thumb, and the wrist. (Weak for)	There is insufficient evidence to recommend for or against constraint-induced movement therapy to improve upper extremity motor outcomes for individuals with some movement in the paretic limb.	Neither for nor against	Reviewed, new-replaced	Supporting (60–62) Additional (63)
12	In patients with motor deficits, there is insufficient evidence to recommend for or against starting a selective serotonin reuptake inhibitor within 30 days of stroke to improve motor recovery and functional outcomes. (Neither for nor against)	There is insufficient evidence to recommend for or against selective serotonin reuptake inhibitors to improve motor outcomes in patients with or without depression.	Neither for nor against	Reviewed, new-replaced	Supporting (64, 65) Additional (66–71)
13	–	There is insufficient evidence to recommend for or against aquatic therapy, as compared with land-based therapy, to improve mobility, balance, and activities of daily living.	Neither for nor against	Reviewed, new-added	Supporting (72)
14	–	There is insufficient evidence to recommend for or against biofeedback as an adjunct intervention to improve motor outcomes.	Neither for nor against	Reviewed, new-added	Supporting (73, 74)
15	–	There is insufficient evidence to recommend for or against motor imagery to improve motor function.	Neither for nor against	Reviewed, new-added	Supporting (75–77)
16	–	There is insufficient evidence to recommend for or against acupuncture to improve motor function.	Neither for nor against	Reviewed, new-added	Supporting (78, 79)
–	We recommend cardiovascular exercise to increase maximum walking speed after stroke. (Strong for)	–	–	–	–
17	We suggest offering functional electrical stimulation, neuromuscular electrical stimulation, or transcutaneous electrical nerve stimulation as an adjunctive treatment to improve upper and lower extremity motor function. (Weak for)	We suggest neuromuscular electrical stimulation to improve motor outcomes.	Weak for	Reviewed, new-replaced	Supporting (80–89)

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Guideline Number	2019 VA/DOD Guideline Recommendation (Strength)	2024 VA/DOD Guideline Recommendation	2024 Strength of Recommendation	2024 Category	2024 References
18	We suggest offering robot-assisted movement therapy as an adjunct to conventional therapy in patients with deficits in upper limb function to improve motor skill. (Weak for) There is insufficient evidence to recommend for or against the use of robotic devices during gait training. (Neither for nor against)	There is insufficient evidence to recommend for or against robot-assisted therapy to improve upper or lower extremity motor outcomes.	Neither for nor against	Reviewed, new-added	Supporting (90–93)
19	We suggest offering virtual reality to enhance gait recovery. (Weak for)	There is insufficient evidence to recommend for or against virtual reality to improve balance or enhance gait recovery.	Neither for nor against	Reviewed, new-replaced	Supporting (94–98)
20	There is insufficient evidence to recommend for or against the use of virtual reality for improving activities of daily living and non-gait motor function. (Neither for nor against)	There is insufficient evidence to recommend for or against the use of virtual reality/serious gaming to improve upper extremity motor outcomes, activities of daily living, or quality of life.	Neither for nor against	Reviewed, new-added	Supporting (91, 99–101)
21	We suggest offering functional electrical stimulation to manage shoulder subluxation. (Weak for)	There is insufficient evidence to recommend for or against contralaterally controlled functional electrical stimulation to improve upper extremity motor outcomes and activities of daily living.	Neither for nor against	Reviewed, new-added	Supporting (102)
22	–	There is insufficient evidence to recommend for or against non-invasive brain-computer interface to improve upper extremity motor outcomes and activities of daily living.	Neither for nor against	Reviewed, new-added	Supporting (91, 103, 104)
23	–	There is insufficient evidence to recommend for or against vagus nerve stimulation as an adjunct intervention for rehabilitation of acute and chronic motor deficits.	Neither for nor against	Reviewed, new-added	Supporting (105)
–	For patients with foot drop, we suggest offering either functional electrical stimulation or traditional ankle foot orthoses to improve gait speed, as both are equally effective. (Weak for)	–	–	–	–
24	We recommend botulinum toxin for patients with focal spasticity that is painful, impairs function, reduces the ability to participate in rehabilitation, or compromises proper positioning or skin care. (Strong for)	We suggest botulinum toxin for patients with focal spasticity depending on patient characteristics and preferences.	Weak for	Reviewed, new-replaced	Supporting (106–111)
25	–	There is insufficient evidence to recommend for or against the use of acupuncture or dry needling for spasticity management.	Neither for nor against	Reviewed, new-added	Supporting (112–115)
26	–	There is insufficient evidence to recommend for or against whole body or localized muscle vibration for spasticity management.	Neither for nor against	Reviewed, new-added	Supporting (116)

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Guideline Number	2019 VA/DOD Guideline Recommendation (Strength)	2024 VA/DOD Guideline Recommendation	2024 Strength of Recommendation	2024 Category	2024 References
27	–	There is insufficient evidence to recommend for or against extracorporeal shock wave therapy for spasticity management.	Neither for nor against	Reviewed, new-added	Supporting (107, 117)
–	We suggest offering intrathecal baclofen treatments for patients with severe chronic lower extremity spasticity that cannot be effectively managed by other interventions. (Weak for)	–	–	–	–
Dysphagia, cognition, and aphasia					
28	We suggest offering Shaker or chin tuck against resistance exercises in addition to conventional dysphagia therapy. (Weak for)	We suggest chin tuck against resistance exercises for patients with dysphagia.	Weak for	Reviewed, new-replaced	Supporting (118–120)
29	We suggest offering expiratory muscle strength training for treatment of dysphagia in patients without a tracheostomy. (Weak for)	We suggest respiratory muscle strength training for dysphagia in patients without a tracheostomy.	Weak for	Reviewed, new-replaced	Supporting (121–123) Additional (123)
30	There is insufficient evidence to recommend for or against tongue to palate resistance training for treatment of dysphagia. (Neither for nor against)	There is insufficient evidence to recommend for or against tongue pressure resistance training for dysphagia.	Neither for nor against	Reviewed, new-replaced	Supporting (124, 125)
31	There is insufficient evidence to recommend for or against neuromuscular electrical stimulation for treatment of dysphagia. (Neither for nor against)	There is insufficient evidence to recommend for or against neuromuscular electrical stimulation and pharyngeal electrical stimulation for dysphagia.	Neither for nor against	Reviewed, new-replaced	Supporting (126–136) Additional (137)
32	There is insufficient evidence to recommend for or against pharyngeal electrical stimulation for treatment of dysphagia. (Neither for nor against)	There is insufficient evidence to recommend for or against surface electromyography for dysphagia.	Neither for nor against	Reviewed, new-added	Supporting (138, 139)
33	–	There is insufficient evidence to recommend for or against the use of selective serotonin reuptake inhibitors to improve cognitive outcomes.	Neither for nor against	Reviewed, new-replaced	Supporting (140–142) Additional (143)
34	–	There is insufficient evidence to recommend for or against computer assisted cognitive rehabilitation to improve cognitive outcomes.	Neither for nor against	Reviewed, new-added	Supporting (144, 145)
35	There is insufficient evidence to recommend for or against the use of intensive language therapy for aphasia (Neither for nor against)	There is insufficient evidence to recommend for or against a specific intensity of language therapy for aphasia.	Neither for nor against	Reviewed, amended	Supporting (146, 147)
36	There is insufficient evidence to recommend for or against hemi field eye patching in addition to traditional therapy for patients with unilateral spatial neglect following stroke. (Neither for nor against)	There is insufficient evidence to recommend for or against hemifield eye patching in addition to traditional therapy to improve functional outcomes in patients with unilateral spatial neglect.	Neither for nor against	Reviewed, new-replaced	Supporting (148–150) Additional (151–154)

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Guideline Number	2019 VA/DOD Guideline Recommendation (Strength)	2024 VA/DOD Guideline Recommendation	2024 Strength of Recommendation	2024 Category	2024 References
37	Among patients with unilateral spatial neglect, there is insufficient evidence to recommend for or against the use of prisms. (Neither for nor against) Among patients with hemianopsia, there is insufficient evidence to recommend for or against the use of prisms or visual search training. (Neither for nor against)	There is insufficient evidence to recommend for or against the use of prism adaptation therapy for patients with unilateral spatial neglect.	Neither for nor against	Reviewed, amended	Supporting (40, 149, 155–158) Additional (39, 159)
–	In patients with dysphagia in the post-acute phase of stroke who require tube feeding, we suggest offering gastrostomy tube over nasogastric tube for maintenance of optimal nutrition. (Weak for)	–	–	–	–
–	There is insufficient evidence to recommend for or against the use of any specific cognitive rehabilitation methodology or pharmacotherapy to improve cognitive outcomes. (Neither for nor against)	–	–	–	–
Mental health					
38	–	There is insufficient evidence to recommend for or against solution-focused psychological interventions (e.g., motivational interviewing, problem-solving therapy) to prevent the development of depression.	Neither for nor against	Reviewed, new-added	Supporting (160) Additional (161)
39	For the prevention of post-stroke depression, there is insufficient evidence for or against the universal use of selective serotonin reuptake inhibitor or a serotonin norepinephrine reuptake inhibitor due to the risk of fractures. (Neither for nor against)	We suggest against the use of antidepressants for the prevention of post-stroke depression.	Weak against	Reviewed, new-replaced	Supporting (162–165) Additional (166, 167)
40	We suggest offering a selective serotonin reuptake inhibitor or a serotonin norepinephrine reuptake inhibitor for treatment of post-stroke depression. (Weak for)	We suggest a selective serotonin reuptake inhibitor or a serotonin norepinephrine reuptake inhibitor for depression symptoms.	Weak for	Reviewed, amended	Supporting (168–172) Additional (161)
41	We suggest offering cognitive behavioral therapy for treatment of post-stroke depression. (Weak for)	We suggest psychotherapy (e.g., cognitive behavioral therapy) for depression following stroke.	Weak for	Reviewed, new-replaced	Supporting (162, 173–179)
42	–	We suggest mindfulness-based therapies for treatment of depression following stroke.	Weak for	Reviewed, new-added	Supporting (180) Additional (181–185)
43	–	There is insufficient evidence to recommend for or against acupuncture, either alone or as an adjunct to pharmacotherapy, for depression following stroke.	Neither for nor against	Reviewed, new-added	Supporting (186–189) Additional (190)

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Guideline Number	2019 VA/DOD Guideline Recommendation (Strength)	2024 VA/DOD Guideline Recommendation	2024 Strength of Recommendation	2024 Category	2024 References
–	There is insufficient evidence to recommend for or against treatment with a combination of pharmacotherapy (selective serotonin reuptake inhibitor/serotonin norepinephrine reuptake inhibitor) and psychotherapy (cognitive behavioral therapy) for treatment of post-stroke depression. (Neither for nor against)	–	–	–	–
–	There is insufficient evidence to recommend for or against pharmacotherapy or psychotherapy for the treatment of post stroke anxiety. (Neither for nor against)	–	–	–	–
–	We suggest offering exercise as adjunctive treatment for post-stroke depression or anxiety symptoms. (Weak for)	–	–	–	–
–	We suggest offering mind-body exercise (e.g., tai chi, yoga, qigong) as adjunctive treatment for post-stroke depression or anxiety symptoms. (Weak for)	–	–	–	–
Telehealth					
44	–	We suggest either face-to-face therapy or telerehabilitation, depending on patient characteristics and preferences.	Weak for	Reviewed, new-added	Supporting (191, 192)
45	–	There is insufficient evidence to recommend for or against the use of telerehabilitation and technology-based interventions to improve stroke-related dysphagia or aphasia outcomes or both.	Neither for nor against	Reviewed, new-added	Supporting (193–197)
46	–	There is insufficient evidence to recommend for or against technology-based caregiver support/education interventions to improve caregiver quality of life.	Neither for nor against	Reviewed, new-added	Supporting (198, 199)
Noninvasive brain stimulation					
47	There is insufficient evidence to recommend for or against the use of transcranial direct current stimulation to improve activities of daily living. (Neither for nor against) There is insufficient evidence to recommend for or against repetitive transcranial magnetic stimulation to improve upper or lower extremity motor function. (Neither for nor against)	There is insufficient evidence to recommend for or against non-invasive brain stimulation (e.g., repetitive transcranial magnetic stimulation, transcranial direct current stimulation, and continuous theta burst stimulation) for patients in stroke rehabilitation.	Neither for nor against	Reviewed, new-replaced	Supporting (40, 41, 200–225)
Other functions					
–	There is insufficient evidence to recommend for or against any specific assessments or interventions regarding return to work. (Neither for nor against)	–	–	–	–

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Guideline Number	2019 VA/DOD Guideline Recommendation (Strength)	2024 VA/DOD Guideline Recommendation	2024 Strength of Recommendation	2024 Category	2024 References
–	There is insufficient evidence to recommend for or against using any specific assessments or interventions to facilitate return to driving. (Neither for nor against)	–	–	–	–

DOD = U.S. Department of Defense; VA = U.S. Department of Veterans Affairs.

ADLs. Task-specific practice involves practice of a whole task or pretask movements for a whole limb or limb segment, such as grasp, grip, or movement in a trajectory to facilitate mobility or ADLs. These movements can include upper- and lower-limb movements, balance activities in a sitting or standing position, transfers, and functional mobility (for example, stairs or household ambulation). The current systematic evidence review found no new studies. The WG considered the assessment of the evidence put forth in the 2019 VA/DOD stroke rehabilitation CPG (24–34) and continues to recommend task-specific practice.

Evidence supports the use of mirror therapy for improvement in motor outcomes and ADLs (change in recommendation from "neither for nor against" in 2019 to "weak for" in 2024; recommendation 6) (**Table**). In mirror therapy, a mirror is placed in front of the patient and the patient is instructed to perform motor activities with the unaffected limb and view the image in the mirror. The patient simultaneously attempts to perform the same activities with the affected limb. It is theorized that visual feedback harnesses neuroplasticity principles because the brain perceives the affected limb as the sound one. In 2 SRs of 62 RCTs ($n = 1982$), mirror therapy provided statistically and clinically significant benefits for several motor outcomes, including motor function, motor impairment, and ADLs (35, 36).

Evidence is equivocal for body-weight support treadmill training to improve motor outcomes (change in recommendation from "weak for" in 2019 to "neither for nor against in 2024"; recommendation 8) (**Table**). Body-weight support treadmill training is a task-specific technique for improving gait. The patient is partially suspended using a body harness to reduce (offload) the relative weight of the patient and provide postural support while the patient is walking on a treadmill. The amount of offloading is gradually decreased as the patient's control of posture and gait improves. One SR included 23 RCTs ($n = 1452$) and found no significant difference between body-weight support treadmill training and conventional overground training for key outcomes (46).

Evidence supports the use of rhythmic auditory stimulation as an adjunct intervention for improvement of motor outcomes (no change in recommendation; "weak for"; recommendation 9) (**Table**). Rhythmic auditory stimulation uses an external rhythm or music to promote improvement in gait or other rhythmic movements. The evidence review included 1 SR of 22 RCTs ($n = 742$) comparing rhythmic auditory stimulation plus control interventions versus control interventions alone (50). The rhythmic auditory stimulation group showed improvement in step cadence, velocity, Fugl–Meyer Assessment, Berg Balance Scale, and overall balance index.

Evidence is equivocal for the following interventions, and therefore recommendations are "neither for nor against": high-intensity interval training over moderate-intensity continuous training to enhance gait recovery (change in recommendation from "strong for" in 2019; the 2019 recommendation was specific only to cardiovascular exercise rather than high-intensity interval training; recommendation 10) (**Table**) and constraint-induced movement therapy for improvement in upper-extremity motor outcomes for persons with some movement in the paretic limb (change in recommendation from "weak for" in 2019; recommendation 11) (**Table**).

Technology-Assisted Physical Rehabilitation

Evidence supports the use of functional electrical stimulation, neuromuscular electrical stimulation, or transcutaneous electrical nerve stimulation for both upper- and lower-extremity motor outcomes (no change in recommendation; "weak for"; recommendation 17) (**Table**) (80–87). These modes of electrical stimulation are used to strengthen and reeducate muscles or muscle groups. Two SRs that together included 46 studies ($n = 1900$) (88, 89) concluded that neuromuscular electrical stimulation triggered or controlled by electromyography was effective in improving upper-limb motor impairment (89). One SR and MA found benefits of neuromuscular electrical stimulation for improvement in ADL function but less clear evidence for improvement in functional motor ability (88). The studies had variable stimulation protocols in pulse, duration, and frequency.

Evidence is equivocal for virtual reality to improve balance or enhance gait recovery (change in recommendation from "weak for" in 2019 to "neither for nor against" in 2024; recommendation 19) (**Table**).

Spasticity

Spasticity is a common poststroke complication and is characterized by abnormal velocity-dependent

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increases in muscle tone and exaggerated reflexes. Spasticity can interfere with function and cause pain but, as is well known in the field of rehabilitation, can also benefit the patient in functional mobility tasks (226).

Evidence supports the use of botulinum toxin for patients with focal spasticity depending on patient characteristics and preferences (change in recommendation from "strong for" in 2019 to "weak for" in 2024; recommendation 24) (Table) (106, 107). A small (n = 29) RCT compared the efficacy of botulinum toxin, 100 to 300 IU (from a predetermined diagram for dosing), versus oral baclofen, 30 to 80 mg/d (in multiple doses titrated to effect and tolerance), on multiple upper- and lower-limb spasticity outcomes, including the Modified Ashworth Scale and Brunnstrom Stages of Stroke Recovery (106). No significant difference was found between botulinum toxin and oral baclofen in any outcome except ankle spasticity, which favored botulinum toxin (treatment difference of 1 point on the Modified Ashworth Scale). Evidence supports botulinum toxin as an effective treatment option for spasticity but also indicates that botulinum toxin may not have a dramatically different place in therapy from oral antispasmodics based on efficacy alone.

Key Area 3: Dysphagia, Aphasia, and Cognition

Dysphagia

Dysphagia is a common sequela of stroke affecting approximately 42% of stroke survivors (227). Unmitigated poststroke dysphagia may cause malnutrition, dehydration, infection, and death. Dysphagia treatment plans targeting impaired oropharyngeal swallowing physiology may be beneficial in the recovery of swallowing function.

Evidence supports that the chin tuck against resistance in addition to conventional dysphagia therapy may improve oropharyngeal swallow function (no change in recommendation; "weak for"; recommendation 28) (Table). Chin tuck against resistance targets suprahyoid musculature, which is key in hyoid and laryngeal movement essential to repositioning the laryngeal entrance to the airway as food and liquid boluses move through the pharynx as well as in traction forces of the upper esophageal sphincter, which allow a bolus to transit into the esophagus. Implementation of chin tuck against resistance resulted in reductions in aspiration in patients with poststroke dysphagia and was also shown to outperform other swallowing exercises (118).

Evidence suggests that respiratory muscle strength training is effective in the treatment of dysphagia for patients without tracheostomy (no change in recommendation; "weak for"; recommendation 29) (Table). Respiratory muscle strength training refers to any number of therapeutic techniques using a device that targets strengthening of the muscles of respiration through resistance for training the actions of inhalation, exhalation, or both. Using respiratory muscle strength training in patients with poststroke dysphagia may decrease aspiration (121), potentially reducing the risk for associated respiratory complications.

Aphasia

Aphasia affects an estimated 30% of stroke survivors (228). Aphasia is an impairment of access to language and affects verbal production, auditory comprehension, written expression, and reading comprehension. Achieving functional communication in poststroke aphasia is an important aspect of recovery and can directly affect quality of life. Aphasia rehabilitation may target specific linguistic functions like word retrieval or syntactic processing, or it may target how a person communicates with others using functional approaches. Although various approaches to aphasia rehabilitation have demonstrated effectiveness (229, 230), the effect of treatment intensity on language recovery is less well understood.

Evidence is equivocal regarding a specific intensity of language therapy for aphasia (no change in recommendation; "neither for nor against"; recommendation 35) (Table). One RCT compared intensive interventions versus usual care in the treatment of chronic poststroke aphasia and found that aphasia severity did not differ significantly between groups, but several secondary outcomes, such as naming, functional communication, and communication-related quality of life, had statistically significant improvements in the higher-intensity groups compared with usual care (146). However, only word retrieval benefits were maintained at the follow-up time point.

Cognitive Impairments

Cognitive impairments affect up to 70% of stroke survivors (231). These impairments are associated with disability, dependency, and morbidity and can present a major challenge to patients, caregivers, and health systems. Yet, effective treatment options remain limited for poststroke cognitive impairment despite the significant societal impact. Evidence is equivocal regarding the use of selective serotonin reuptake inhibitors (SSRIs) to improve cognitive outcomes (no change in recommendation; "neither for nor against"; recommendation 33) (Table). An RCT compared the effect of escitalopram versus placebo on cognitive function, as measured by the Mini-Mental State Examination. No statistically significant differences were found between groups (140). In addition, the Mini-Mental State Examination is not a comprehensive measure of cognitive functioning because it is meant to be used as a screening tool to identify whether further assessment is warranted. Therefore, the WG did not consider this to be an ideal outcome measure for cognitive function.

Unilateral Spatial Neglect

Unilateral spatial neglect may occur in two thirds of patients with acute right hemisphere stroke (232). Patients are unaware of and do not respond to stimuli present on the side opposite the brain lesion. Traditional therapy approaches include strategies to direct attention

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toward the side of neglect. Evidence supports the use of mirror therapy for the treatment of unilateral spatial neglect (new recommendation; "weak for"; recommendation 7) (**Table**). One SR of 4 RCTs ($n = 214$) examined the effect of mirror therapy on unilateral spatial neglect and found a large positive effect on ADL performance (41).

Key Area 4: Mental Health

Treatment of Poststroke Depression

Evidence supports the use of the following interventions for the treatment of poststroke depression (recommendations all unchanged; "weak for"): SSRIs or serotonin–norepinephrine reuptake inhibitors (SNRIs) (recommendation 40) (**Table**) (168–170), psychotherapy (such as cognitive behavioral therapy [CBT]) (recommendation 41) (**Table**) (162, 173), and mindfulness-based therapies (recommendation 42) (**Table**) (180).

Three SR and MAs found that SSRIs and SNRIs consistently outperformed placebo (or "routine treatment") on depression scores as measured by the Hamilton Depression Rating Scale at the end of treatment and at longer-term follow-up (168–170). The WG considered the risks of untreated depression, which is associated with increased disability and death in persons with stroke (161), as well as the known risks of SSRI or SNRI therapy and concluded that for persons who have known depression, the benefits of therapy exceed the risks.

Cognitive behavioral therapy is a class of interventions designed to change dysfunctional forms of thinking and behavior to directly reduce psychological suffering (173). Two SR and MAs showed that CBT and other behavioral therapies improved general depression symptoms (162, 173). One SR and MA of 10 studies (6 of which were RCTs) ($n = 672$) in which the intervention groups received computer-based CBT or traditional CBT and the control groups received various alternative behavioral therapies showed that CBT improved general depression symptoms immediately after the intervention and at 3-month follow-up (173). A Cochrane review of 6 RCTs ($n = 521$) examined various psychological therapies, including structured CBT, compared with usual care or attention control. The psychological therapies, including CBT, resulted in improvement in depression prevalence and depression symptoms at the end of treatment (162).

Mindfulness is a process of being fully present in the moment with intention and a nonjudgmental awareness. One SR and MA of 7 RCTs ($n = 502$) evaluated the effectiveness of mindfulness-based stress reduction and mindfulness-based cognitive therapy for poststroke depression (180). Comparator groups received usual care, education, or no treatment. Self-reported depression scores were significantly lower in the groups that received mindfulness-based stress reduction or mindfulness-based cognitive therapy.

The evidence was equivocal regarding the use of acupuncture, either alone or as an adjunct to pharmacotherapy, for depression after stroke (new recommendation; "neither for nor against"; recommendation 43) (**Table**) (186–188). Acupuncture is a traditional Chinese medicine technique done by highly trained practitioners using specialized thin needles for the treatment of multiple health conditions, including depression. The evidence about acupuncture for the treatment of poststroke depression was mixed and did not show clear benefit (186–188). The WG also considered the 2022 VA/DOD Clinical Practice Guideline for the Treatment of Major Depressive Disorder, which recommended neither for nor against acupuncture as an adjunct for the treatment of major depressive disorder (233). The WG concluded that no recommendation could be made for or against acupuncture.

Prevention of Poststroke Depression

As many as half of stroke survivors might develop depression within 5 years of their stroke (161). Evidence was equivocal regarding the use of solution-focused psychological interventions (such as motivational interviewing and problem-solving therapy) for the prevention of poststroke depression (new recommendation; "neither for nor against"; recommendation 38) (**Table**). The goal of solution-focused psychological interventions is to identify problems and barriers hindering patients' recovery, thereby empowering them to make progress toward their goals. A Cochrane review of 7 RCTs ($n = 607$) comparing these types of interventions versus usual care or attention control found that, although the proportion of people who developed depression was reduced, the significant methodological concerns in the studies precluded any firm conclusions (160).

Evidence in combination with risk–benefit assessment led the WG to suggest against the use of antidepressants for the prevention of poststroke depression (change in recommendation from "neither for nor against" in 2019 to "weak against" in 2024; recommendation 39) (**Table**). A Cochrane review of 12 RCTs ($n = 734$) examined various antidepressants for this purpose, including SSRIs and SNRIs, compared with placebo (162). Patients in the experimental group were less likely to develop depression; however, the severity of depressive symptoms did not differ between groups. The FOCUS (Fluoxetine Or Control Under Supervision) trial (2019; $n = 3127$) showed a doubling of the risk for bone fractures in patients treated with fluoxetine (171). The WG believed that, given the potential risk for bone fractures and the complications of polypharmacy in a primarily elderly population, the benefits of long-term use of antidepressants might not outweigh the risks, especially when antidepressants are prescribed for the prevention of a future illness that might not even occur.

COMPARISON WITH OTHER GUIDELINES

This evidence-based guideline uniquely offers the most up-to-date review of the management of common stroke sequelae and the systems and processes for rehabilitation. It is in contrast to the guidelines from

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the American Heart Association and American Stroke Association on the early management of acute ischemic stroke, secondary prevention of stroke, and management of spontaneous intracerebral hemorrhage, which are focused on the acute management of ischemic and hemorrhagic stroke as well as secondary stroke prevention (234–236)—also vital stages of the stroke care continuum. The more closely related Guidelines for Adult Stroke Rehabilitation and Recovery from the American Heart Association and American Stroke Association were last published in 2016 (237). Although they also focused on management of stroke sequelae, their scope was broader and included system-level considerations, prevention and medical management of comorbid conditions, sequelae assessments and tools, sensorimotor treatments or interventions, and transitions of care and community rehabilitation (237). Most of those guidelines have formed current practice and standards of care. Our guidelines have benefited from these prior publications and sought to continue to propel the best clinical practices in adult stroke rehabilitation and guide future research.

RESEARCH RECOMMENDATIONS

The evidence base in stroke rehabilitation was limited by known methodological biases and challenges in conducting high-quality research in rehabilitation medicine (238, 239), as well as the heterogeneous nature of strokes and the patients they affect. Specifically, the introduction of bias into rehabilitation research is unavoidable in most studies for several reasons. First, the stroke population is heterogeneous (and attempts to make it more homogeneous would markedly limit study generalizability). Second, recruitment can be limited because of patients' fears they will be put in the control group. Third, blinding of participants to the rehabilitation intervention is sometimes impossible. Finally, control groups are often "usual care" rehabilitation, which varies from patient to patient and from day to day. Future research should consider study designs, power, randomization, blinding, and standardization of interventions and outcome measures. This review also identified more specific gaps. We suggest increased use of the Rehabilitation Treatment Specification System in poststroke motor intervention research (240). More research on the use of robotics, combination therapies, and other advanced techniques in rehabilitation is also needed. When designing studies to investigate mental health outcomes, researchers should standardize their inclusion and exclusion criteria on the basis of clinically relevant and accepted diagnostic criteria. Furthermore, although telemedicine and technology-based interventions have become more widely available, there are limited data on the optimal level of support these interventions should provide to patients, the appropriate degree of provider interaction, and which outcomes are most beneficial. We recommend incorporating these care delivery models when feasible or as a component of noninferiority studies. Finally, patient adherence needs to be well defined, standardized, and measured to lend insight into the effectiveness of interventions. With the exciting growth of modern medical and technological advances, additional specific research recommendations can be found in the full guideline (www.healthquality.va.gov/guidelines/rehab/stroke/index.asp) (5).

CONCLUSION

The 2024 VA/DOD CPG on stroke management provides recommendations and clinical algorithms for poststroke rehabilitation. This article summarizes where evidence is strongest to support guidelines on key areas most relevant for primary care physicians: transitions to community (case management and psychosocial interventions); motor therapy (task-specific practice, mirror therapy, rhythmic auditory stimulation, electrical stimulation, and botulinum toxin for spasticity); dysphagia, aphasia, and cognition (chin tuck against resistance and respiratory muscle strength training); and mental health (SSRI use, psychotherapy, and mindfulness-based therapies for treatment but not prevention of depression). Poststroke rehabilitation requires an interdisciplinary, holistic approach to the management, treatment, and rehabilitation of stroke sequelae, with the patient and caregivers as vital members of the team. We recommend using these guidelines as an adjunct to the Guidelines for Adult Stroke Rehabilitation and Recovery from the American Heart Association and American Stroke Association (237).

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